CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

3^{rd} Semester of B. Tech. (EC) Examination

Nov- Dec 2012

EE221 Electrical Technology

Date: 01.12.2012, Saturday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Explain the following parts of D.C machine : (i) Yoke (ii) Pole core and pole shoe (iii) Armature core (iv) Commutator (v) Brushes and bearings **[07]**

Q - 2 (a) Explain armature torque and shaft torque of a D.C motor. **[04]**

(b) Explain the characteristics of D.C shunt motor. **[05]**

(c) A shunt generator has a full load current of 196A at 220V. The stray losses are 720W and the shunt field coil resistance is 55Ω . If it has full load efficiency of 88%, find the armature resistance. Also, find the load current corresponding to maximum efficiency. **[05]**

OR

Q - 2 (a) Derive the EMF equation of a single phase transformer. **[04]**

(b) A 4-pole, 240V, wave connected shunt motor gives 1119 kW when running at 1000 r.p.m and drawing armature and field currents of 50A and 1.0A respectively. It has 540 conductors. Its resistance is 0.1Ω. Assuming a drop of 1 volt per brush, find (a) Total torque (b) Useful torque (c) Useful flux per pole (d) Rotational losses (e) Efficiency **[05]**

(c) Explain the operation of single phase transformer on no- load condition with the help of vector diagram. **[05]**

Q - 3 (a) Explain the relationship between torque and slip in an induction motor with the help of suitable graph. **[04]**

(b) A 50 kVA, 4400/220 V transformer has R_1 = 3.45Ω , R_2 = 0.009Ω. The values of reactances are X_1 = 5.2Ω and X_2 = 0.015Ω. Calculate for the transformer (i) Equivalent resistance as referred to primary (ii) Equivalent resistance as referred to secondary (iii) Equivalent reactance as referred to both primary and secondary (iv) Equivalent impedance as referred to both primary and secondary (v) Total copper **[06]** loss, first using individual resistances of the two windings and secondly, using equivalent resistances as referred to each side.

(c) Why does the rotor rotate in a three-phase induction motor? [04]

OR

Q - 3 (a) Draw and explain equivalent circuit of a single phase transformer along with necessary equations. [07]

(b) A 440V, 3-phase, 4-pole,50Hz, star connected induction motor has a full load speed of 1425 r.p.m. The rotor has impedance of (0.4+j4) Ω and rotor/stator turn ratio of 0.8. Calculate (i) Full load torque (ii) Rotor current and full load rotor copper loss (iii) Power output if windage and friction losses amount to 500W (iv) maximum torque and the speed at which it occurs (v) Starting current (vi) Starting torque. [07]

SECTION – II

Q - 4 (a) Explain the production of rotating field in an induction motor for 2 phase supply. [07]

Q - 5 (a) Explain shaded pole single phase induction motor. [07]

(b) A 100 kVA, 3000V, 50Hz 3-phase star connected alternator has effective armature resistance of 0.2Ω. The field current of 40A produces short circuit current of 200A and an open circuit emf of 1040V (line value). Calculate the full load voltage regulation at 0.8 p.f lagging and 0.8 p.f leading. [07]

OR

Q - 5 (a) Explain the double field revolving theory of a single phase induction motor. [07]

(b) Derive the EMF equation of alternator. [05]

(c) Explain the terms : (a) Pitch factor (b) Distribution factor [02]

Q - 6 (a) Explain different types of equipments used in substation along with their functions. [07]

(b) Explain Transformer substation with the help of its block diagram. [07]

OR

Q - 6 (a) Explain any two techniques of Power factor Improvement. [06]

(b) Explain Single bus-bar arrangement system in a sub-station with neat diagram. [05]

(c) Write a comparison of indoor and outdoor substation. [03]

****